

TEMS Automatic Administration Guide

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1. General

This document describes recommendations for how you can administer Microsoft SQL Server 2005 database server used by TEMS Automatic. The document covers the following topics:

- How to configure database and logfile backup routines. Limitations and consequences of various backup strategies are discussed.
- How to restore the system to a known state so that it can be restarted.
- How to administer database users.
- How to administer logfile storage databases.
- What admin tasks to perform on a daily, weekly, and monthly basis.

The installation of the database server is described in ref [1], TEMS Automatic Database Server Installation Guide.

For information on dimensioning the Measurement and Statistics databases and OLAP Server Geobases, see ref [2], TEMS Automatic System Configuration and Dimensioning.

Regarding system requirements and system configuration, please refer to the relevant documents in the user documentation.

2. Backup and Recovery

This chapter describes a backup strategy which enables the recovery of a TEMS Automatic system with a minimum of data loss. How to perform the recovery is explained as well.

Provided that these procedures are rigorously implemented, the worst-case data loss in case of system failure is one day's worth of measurements. The data loss can be further reduced by reprocessing of the most recent logfiles.

General Note: Many of the database crashes reported by TEMS Automatic customers have been caused by disk failure. The standard Raid-5 configuration allows one disk to fail without the databases being affected. It is when the second disk fails that the database problem occurs. It is therefore recommended to monitor the state of the Raid-5 disks to catch the first disk failure.

2.1. Backup

This is the recommended backup procedure. It can be summarized as a *weekly full backup* of TEMS Automatics Measurement and Statistics database combined with *daily differential database backups* and *transaction log backup* each hour for both measurement and statistics database. The order of the backup jobs is critical if the system needs to be restored this is the sequence to follow. First, before any daily or weekly backup of the TEMS Automatic Measurement or Statistics databases are performed the daily processing of statistics needs to be completed and the TLH service needs to be stopped. After these prerequisites have been met a weekly or daily backup of the TEMS Automatic databases can be performed.

The recommended recovery model is bulk logged both for the measurement and the statistics database.

The storage of the backups can be critical in cases of fire or theft and the recommendation is to store the previous month's backup in a different location.

How to handle backups in practice is covered in chapter 4. Backup of folders, being trivial, is not discussed further.

Software backups are not needed, since both TEMS Automatic components and third-party software can be reinstalled from installation media.

2.1.1. Weekly Backups

A *weekly full backup* to disk or tape should be made of:

- **TEMS Automatic Measurement** database
- **TEMS Automatic Statistics** database.
- **TEMS Automatic Logfile Storage** database, if used.

- The **OLAP Server Geobases**. The archiving of Geobases should likewise be automated; see section 4.2.
- The *logfiles* and *TU administration files* in the **Region\Converted folders**. The folders should be emptied upon backup to keep their size down.
- The *map files* in the **Map Server folder**.

2.1.2. Daily Backups

The following backups should be done each day:

- Differential backup of **TEMS Automatic Measurement** database
- Backup to separate disk of the **Region\Converted folders**. Activate the backup option in the *ComServer* application to take care of this automatically. Use the backup disk to recover data that has not been stored on disk or tape during the weekly backup (see section 2.1.1 above).

Additionally, the *msdb* database should be backed up after every backup operation to make the restore procedure as easy as possible. The *msdb* database contains the backup history; it also stores the SQL Server Agent Jobs for archiving Geobases.

There is no transaction log backup function in the OLAP Server, so for Geobases the full backup is the only available option. When recovering the system, the Geobases restored from backup must therefore be incrementally processed to include the most recent data (see section 2.3.4).

2.1.3. Other Backup Tasks

The *master* database has to be backed up whenever changes to logins have been made.

2.2. Backup Event Schedule Example

Below in figure 1 is an example schedule which implements the backup procedure given in section 2.1. It begins with a new system installation. The weekly full backup in this case is done on Saturdays.

Day No.	Day of the Week	Events	Backup Tasks
1	Thursday	New system installation.	Full backup of Measurement, Statistics, Geobases, msdb, and master Backup of Map Server folder.
2	Friday		Differential database backup of Measurement database. Full backup of msdb.
3	Saturday		Full backup of Measurement, Statistics, Geobases, and msdb. Backup of Region\Converted and Map Server directories.
4	Sunday		Differential database backup of Measurement database. Full backup of msdb.

5	Monday		Differential database backup of Measurement database. Full backup of msdb.
6	Tuesday	A new login is created (master database modified).	Differential database backup of Measurement database. Full backup of master. Full backup of msdb.
7	Wednesday		Differential database backup of Measurement database. Full backup of msdb.
8	Thursday		Differential database backup of Measurement database. Full backup of msdb.
9	Friday	New Geobases are created.	Differential database backup of Measurement database. Full backup of msdb.
10	Saturday		Full backup of Measurement, Statistics, Geobases, and msdb. Backup of Region\Converted and Map Server folders.
11	Sunday		Differential database backup of Measurement database. Full backup of msdb.
12	Monday		Differential database backup of Measurement database. Full backup of msdb.
13	Tuesday		Differential database backup of Measurement database. Full backup of msdb.
14	Wednesday		Differential database backup of Measurement database. Full backup of msdb.

Figure 1 Table 2:1

2.2.1. Notes on Backup Event Schedule

References are to the table in section 2.2 figure 1.

- A. Perform a full backup after a new system installation or after a system upgrade. Use the maintenance plan feature described in section 4.1. Create one plan for each database.
- B. Differential database backup of the Measurement database. Store the backup file on another disk than the database, or use a tape drive. Use the maintenance plan feature described in section 4.1.
- C. The weekly full backup of all databases. Use the maintenance plan feature described in section 4.1.
- D. After changes to the master database (e.g. new login, server-wide configuration changes), make a new full backup of the master database.
- E. Schedule the creation and the first full processing of new Geobases just before a full backup of the Statistics database (which contains the Geobases).

2.3. Recovery

The key to successfully restore a crashed system is to have a consistent and recent backup of all data and to combine this with a recovery procedure that will meet the demands of the recovery operation. To meet the demands of a recovery operation there must be an estimation of much time and additional resources different recovery cases will utilize.

For example a disk crash or broken raid controller might corrupt the data in the database. To recover in this case we will need to apply the latest full database backup and additional differential database backups. To meet the demands to recover from the crash within 2 hours we need to know how much time is needed to install a new disk or raid controller, an estimation, based on recovery tests, of how long time it will take to restore the database backups.

Suppose that the TEMS Automatic system crashes on the day following day 14 in figure 1 in section 2.2. The table from that section is continued below in figure 2.

Day No.	Day of the Week	Events	Backup/Recovery Tasks
15	Thursday	<i>Database crash</i>	One or several databases crashes and the TEMS Automatic system recovered from backups. See sections 2.3.1–2.3.4 below.
16	Friday		Differential database backup of Measurement database. Full backup of msdb. Ordinary system operation is resumed.

Figure 2 Table 2:2

If several databases have crashed at the same time they should be restored in the following order:

1. master database
2. msdb database
3. TEMS Automatic Measurement database
4. TEMS Automatic Statistics database.
5. Geobases

In the following section general instructions will be given on how to restore each database. Please review the [Microsoft SQL Server 2005 Books Online](#) for further information.

See chapter 4 for specifics on how to perform various tasks with SQL Server admin tools.

2.3.1. master Database

1. Restore the database from the latest backup; see section 4.6, “Restoring a Database Manually”.

2. If the SQL Server is not accessible, see ref [5], chapter "Restoring the Master Database".
3. If the master database backup is from an older backup set, any logins added after the backup have to be manually added to the database server and connected to the users of the TEMS Automatic databases; see section 4.10, "Connecting Database Users and Logins". Server-wide settings are also stored in the master database, so these settings also have to be manually added if they are not contained in the backup set; see sections 4.11, "Creating a Linked Server" and 4.12, "Changing the Remote Query Timeout Value".

2.3.2. msdb Database

1. Restore the database from the latest backup; see section 4.6, "Restoring a Database Manually".
2. If this is not possible, any DTS package, SQL Server Agent Jobs, and Maintenance plans added after creation of the backup set must be added manually to the database server.

Note: If other databases also need to be recovered from backups, and the msdb is old, the backup history cannot be used as a basis for any restore operations.

2.3.3. Measurement Database

1. Try to make a backup of the active transaction log. Execute the following statement from SQL Server Management Studio to backup the transaction log file without truncating it.

```
BACKUP LOG MeasurementDatabase
TO DISK = 'E:\ MeasurementDatabase Active_log1'
WITH NO_TRUNCATE
GO
```

2. Then restore the Measurement database based on the backup history: see section 4.3, "Restoring a Database using the Maintenance Plan".
 - If the backup of the active transaction log was successful, it should turn up as the last transaction log in the backup history list. In this case, you do not need to reprocess any logfiles or event history files.
 - If the backup of the active transaction log failed, you can reprocess the most recent logfiles and event history files from the Region\Converted folders, that is, those that have a timestamp more recent than the latest backup file/transaction log. (The timestamp is contained in the file name.) Do *not* reprocess older logfiles, since this would cause data redundancy.
3. Restore the Map Server folder from the same backup set as the Measurement database.

2.3.4. Statistics Database and Geobases

If the Statistics database has crashed:

1. Restore the database from the latest Statistics backup set: see section 4.6, "Restoring a Database Manually".
2. Also restore any active Geobases from the same backup set: see section 4.4, "Restoring Geobase". Do incremental processing on the Geobases to bring the Statistics subsystem up to date. (For a crashed Geobase that was not active, no incremental processing is necessary.)
3. If the crashed Geobases are small, consider replacing them by creating new ones.
4. After restoring the crashed databases, make a full backup of the databases.

2.4. Recovery from Older Backups (Data Dependencies)

If the backup procedure in section 2.1 is not followed completely or fails at some point, but older backups exist, old data must be used when recovering the system. The following must then be kept in mind:

- If database backups are not performed (for whatever reason) for a certain period of time, there is a need to reprocess logfiles (*.log) and TU event history files (*.ent) from the Region\Converted folder to be able to restore the system. During recovery, the contents of Region\Converted are placed in the Region\Linehandler\In folder. As in the ordinary case (section 0), you should use *only* files having a timestamp more recent than the latest database backups.
- The Statistics database and the Geobases on the OLAP Server must be from the same backup set. This is because the Geobase configuration settings and dimension tables are stored in the Statistics database.
- The Measurement database must be from the same backup set as the Statistics/Geobase data, or it must be more recent. Measurement data cannot be older than the Geobases, mainly because the Geobases would then contain information that has no corresponding logfile data in the Measurement database: logfile data identities (i.e. values of DataSourceID) would not correspond to valid logfiles. The same is true of any new TUs, TU groups, and areas added to the system after backup of the Measurement database. Moreover, the Geobase configuration data is stored in the Measurement database as well, so if a new Geobase is created after backing up the Measurement database, inconsistency could result.
- If the Statistics database is to be restored to an old state, any Geobase configuration information more recent than this in the Measurement database must be manually removed from the latter, again to avoid data inconsistency.
- The maps used in the Presentation application are stored on the common network drive Map Server. The maps belong to the same backup set as the Measurement database, since their existence is registered there. If an old backup of the Map Server folder is used anyway, and new maps have since been added, some presentations will indicate when opened that one or more maps are missing. If this happens, either remove the map references from the

presentations (the rest of the presentations will still be valid) or add the missing map.

- If the Measurement database is not backed up regularly, it is a very good idea to at least make copies of the tCallEvents table which stores (among other things) the uplink AQM values generated in voice test calls. Since the uplink AQM data is logged on the fixed side, it is not contained in logfiles from test units, and consequently the data is lost if the Measurement database crashes without any sort of backup being available. One more action should be taken in this situation: By default, when a logfile is stored in the Measurement database, the uplink AQM values in tCallEvents are moved by the Logfile Handler to other tables (tCDMAPesqUL or tUMTSPesqUL) and are removed from tCallEvents. This deletion can be prevented by setting the column tConfigTLH.DeleteCallEventRecords in the Measurement database to zero. The tCallEvents table will then keep the uplink AQM data indefinitely, and it suffices to export it as a text file every now and then.

2.5. Recovering the System in the absence of Backups

If no backups are available, proceed as follows in order to get the system up and running again.

2.5.1. Total System Failure

This recovery description is based on the assumption of a more or less total system failure (caused by fire, theft, or the like).

1. If the TUs are still working, they will continue trying to upload logfiles and administration information. Among the subsystems, reinstall the FTP server first (unless it is still working). Once the FTP server is up and running, TUs will be able to perform all their tasks, except that MTUs cannot make voice test calls since these require a Call Generator.
2. Install all third-party software on the server machines (the ones where the *TEMS Automatic databases*, *ComServer*, *Operator Console*, *Call Generator* and *Logfile Handler* will be running).
3. Install the Measurement database.
4. Install the ComServer, Call Generator, and Operator Console.
5. Start the ComServer and let the TUs re-register in the system.
6. Create new work orders for all TUs. Do this even if no changes are needed to the work orders that were used before the system failure. The data collection functionality of the system is now restored.
7. Install the Logfile Handler.
8. Install the Statistics database, the OLAP Server, and the Statistics subsystem.
9. Create new Geobases and process them in order to verify that the system has recovered its post-processing functionality.
10. Install the Presentation application. Verify that the system has been recovered by viewing routes and statistics.

2.5.2. Partial System Failure

This section addresses various situations where some parts of the system survived the crash undamaged. Components not in the headings are assumed to have been damaged, unless otherwise indicated.

2.5.2.1. Measurement Database Undamaged

Recover the rest of the system based on the old Measurement database, i.e. skip step 3 in section 2.5.1 above. The only measurement data that will be lost is the logfiles in the Regions\Logfiles folder that have not yet been stored in the Measurement database. There may be some work orders in the Operator Console that will never be confirmed and must be resent to the TUs.

Before reinstalling the Statistics database structure and recreating the Geobases, remove the old Geobase configuration data that is stored in the Measurement database.

2.5.2.2. Region\Converted and Region\Logfiles Folders Undamaged

The TU logfiles (*.log) and event history files (*.ent) can be reprocessed into the Measurement database after the TUs have reregistered in the system (but *not* before that). During recovery, the contents of the Region\Converted folder are placed in the Region\Linehandler\In folder.

If the Measurement database too is undamaged, and the system recovery is based on the old Measurement database, do *not* reprocess the Regions\Converted folder (since this would cause duplication of data).

2.5.2.3. Statistics Database Undamaged

If the Statistics database is undamaged, it still needs to be reinstalled since the Geobases statistics are linked to the measurement data in the Measurement database (which has been lost).

On the other hand, it is still a good idea to keep the old Geobases. Even if they cannot be hooked up to the reinstalled TEMS Automatic system, you can still access them through other applications, such as Microsoft Excel, and extract their data.

2.5.3. Geobase Crash

If some or all of the Geobases have been damaged, but the rest of the system works, the Geobases can be replaced by creating new ones, without reinstalling any part of the system.

3. User Administration

In the Measurement and Statistics databases a number of roles are defined to give each login the correct level of access to the selected tool or application.

Tool	User Profile	Database Roles
Presentation	Normal user	TEMS_Presentation_User
Operator Console	Administrator user	TEMS_Automatic_Administrator
Operator Console	Normal user	TEMS_Automatic_Operator
Operator Console	Read-only user	TEMS_Automatic_Monitor
Call Generator	Application	TEMS_Automatic_CallGenerator
ComServer	Application	TEMS_Automatic_ComServer
Logfile Handler (TLH)	Application	TEMS_Automatic_TLH
Service Monitor	Application	TEMS_Automatic_Administrator
Database Clean	(no profile exists)	(use database owner, "tems")
Statistics	(no profile exists)	(use database owner, "tems")
AutoDBClean	(no profile exists)	(use database owner, "tems")

The database roles fall into two categories:

- Roles controlling access rights for *normal users*

This group consists of the users of the Operator Console and Presentation applications. These users are straightforward to create and use; see section 4.7, "Adding a User", section 4.8, "Changing User Passwords", and section 4.9, "Altering User Rights".

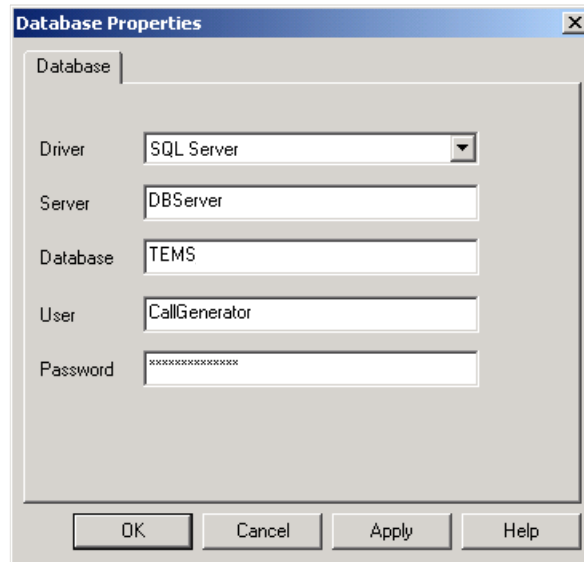
- Roles controlling access rights for *system background applications*

This group consists of the ComServer, Call Generator, Logfile Handler, Service Monitor, and Statistics applications. These users are created at installation time, either by the installation scripts (Logfile Handler, Service Monitor, AutoDBClean), by the installation staff (Call Generator, ComServer), or at first use (Statistics). To modify a database user or create a new one, the following general procedure must be followed: stop the relevant application; make the desired changes to users in the database; reconfigure the application with the new user and password information; finally, restart the application. The details of performing this task are a little bit different for each application and are explained in subsections below.

3.1. Call Generator

1. Open the Call Generator user interface by double-clicking the Call Generator icon on the taskbar.
2. Stop the Call Generator by choosing Action → Stop.
3. Modify the set of database users as desired. For more information on the procedures, see section 4.7, "Adding a User", and section 4.8, "Changing User Passwords".

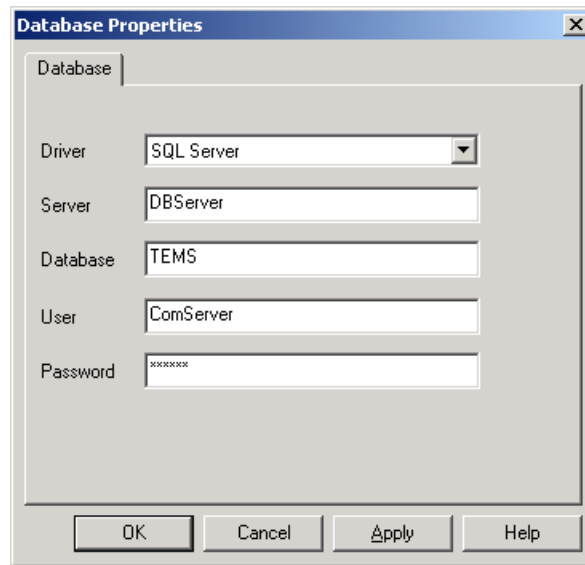
4. In the Call Generator user interface, open the Database dialog by choosing File → Database.



5. Enter the new user and password. Click OK.
6. Start the Call Generator by choosing Action → Start.
7. Verify that the Call Generator is accessing the database by checking that the Call Generator is making test calls (the number of MO/MT calls is increasing).

3.2. ComServer

1. Open the ComServer user interface by double-clicking the ComServer icon on the taskbar.
2. Stop the ComServer by choosing Action → Stop.
3. Modify the set of database users as desired. For more information on the procedures, see section 4.7, "Adding a User", and section 4.8, "Changing User Passwords".
4. Log off the database by choosing File → Log off.
5. In the ComServer user interface, open the Database dialog by choosing File → Database.



6. Enter the new user and password. Click OK.
7. Log on to the database by choosing File → Log on.
8. Start the ComServer by choosing Action → Start.
9. Verify that the ComServer is accessing the database by checking that the ComServer is processing files (the number of converted files is increasing).

3.3. Logfile Handler

The Logfile Handler application does not have a graphical user interface. The user and password are stored in the file LogfileHandler.ini in the application folder.

1. From the Control Panel, open Administrative Tools → Services.
2. Stop the TEMS Logfile Handler Monitor service.
3. Stop the TEMS Logfile Handler service.
4. Modify the set of database users as desired. For additional information on the procedures, see section 4.7, "Adding a User", and section 4.8, "Changing User Passwords".
5. Open the LogfileHandler.ini file. It is usually found under
C:\Program Files\Ericsson\TEMS Automatic\<application name>.
6. In the [Database] section, change the Pwd tag, and set the Crypt tag to zero.

```
[Database]
Driver=SQL Server
Server=SQLServer2000
Database=TEMS_Total
User=LogfileHandler
Pwd=newpassword
Crypt=0
```

7. Modify the User tag if needed.
8. Save and close the LogfileHandler.ini file.
9. Start the TEMS Logfile Handler service.
10. Start the TEMS Logfile Handler Monitor service.
11. Verify that the Logfile Handler is accessing the database by checking that the Logfile Handler is processing logfiles.

3.4. Service Monitor

The Service Monitor application does not have a graphical user interface. The user and password are stored in the file ServiceMonitordr.ini in the application folder. The Service Monitor application exists for the ComServer, Call Generator and Logfile Handler. There is only one Service Monitor for each group of Logfile Handlers and machine, monitoring all Logfile Handlers on that machine that is serving one Measurement database. The same is true for each group of ComServers.

1. From the Control Panel, open Administrative Tools → Services.
2. Stop the relevant service (TEMS ComServer Monitor, TEMS Logfile Handler Monitor or TEMS Call Generator Monitor).
3. Modify the set of database users as desired. For additional information on the procedures, see section 4.7, "Adding a User", and section 4.8, "Changing User Passwords".
4. Open the ServiceMonitordr.ini file. It is usually found under C:\Program Files\Ericsson\TEMS Automatic\<application name>
5. In the [Database] section, change the Password tag. Set the Crypt tag to zero.

```
[Database]
Datasource=DSN=ServiceMonitor;DATABASE=TEMS_Total
User=ServiceMonitor
Password=newpassword
Crypt=0
```

6. Modify the User tag if needed.
7. Save and close the ServiceMonitordr.ini file. Then start the service.

3.5. Statistics

The Statistics application user is the same as the database owner, and some special handling is therefore required to change the password. It is not possible to create a new user or alter the user rights for the database owner with regard to the Statistics application.

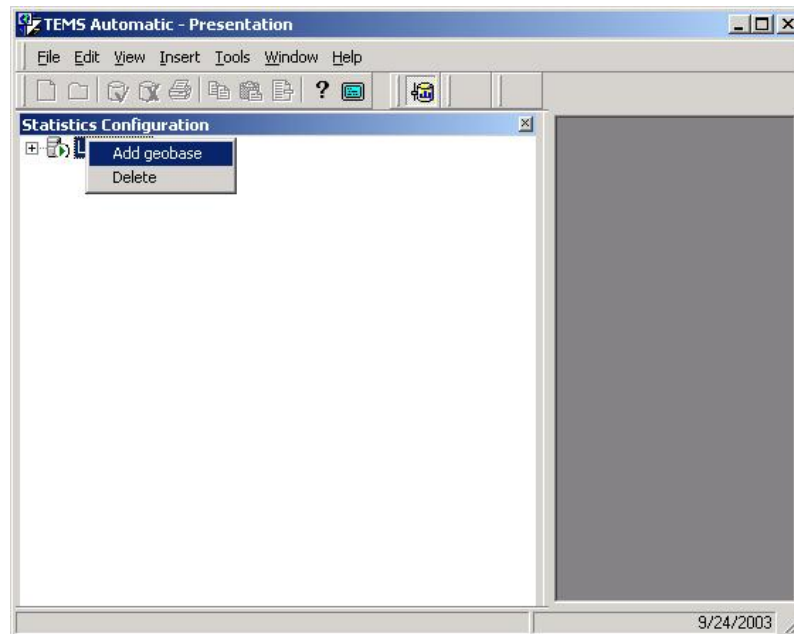
The Statistics application accesses both the Measurement database and the Statistics database. The user and password for the Measurement database are stored in the table MeasDB in the Statistics database when the Statistics application is used for the first time. On later occasions, these login credentials are retrieved automatically and will not be seen any more by the end user. To change the password, the record in MeasDB has to be deleted, whereupon the Statistics application can be started again. The application will ask for the Measurement database parameters; see section 3.5.1 below.

The user and password for the Statistics database are entered every time a new Geobase is created. For existing Geobases the password is stored in the Data source of each Geobase. See section 3.5.2 below.

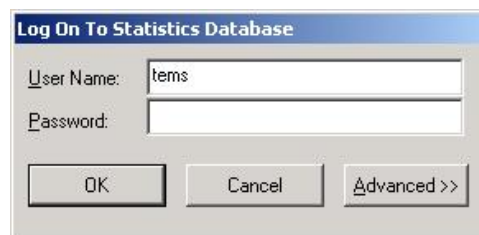
3.5.1. Measurement Database

1. Stop the Statistics application and any scheduled Geobase processing tasks that are currently running. This is necessary in order to avoid doing any processing while the passwords are being updated. Disable any scheduled Geobase processing tasks that are to begin over the next 10–20 minutes.
2. Stop any other Measurement database processes, using the database owner user ("tems").
3. Change the password for the existing Measurement database user (= database owner). See section 4.8, "Changing User Passwords".
4. From the Start menu, open Programs → Microsoft SQL Server 2005 → Sql Server Management Studio.
5. Log on to the Statistics database as database owner ("tems").
6. Run the following SQL statement:

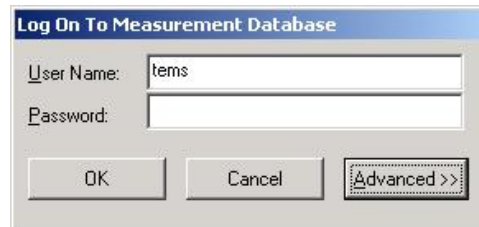

```
DELETE MeasDB
```
7. Start the Presentation application.



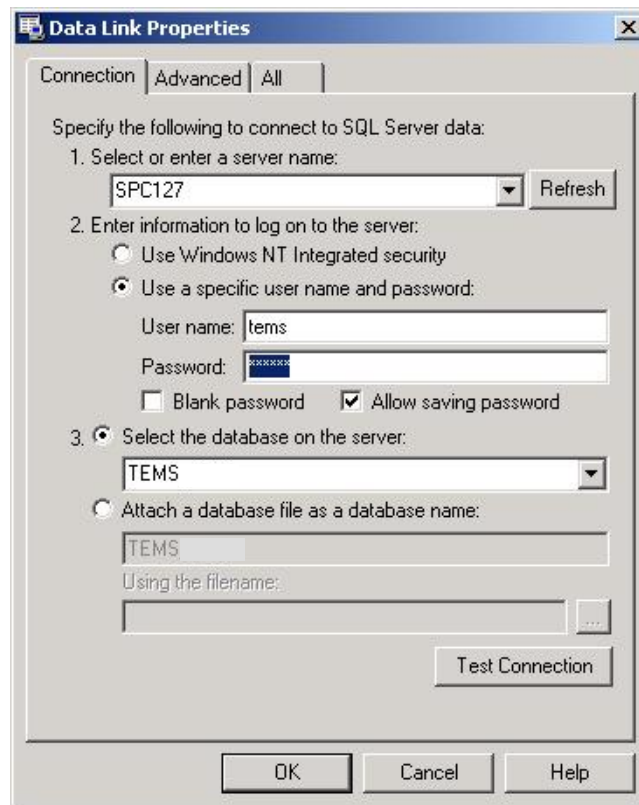
8. Right-click on the OLAP Server node and select Add Geobase.



9. Enter the user (= database owner) and password for the Statistics database, then click OK.



10. In the Log On window of the Measurement database, click the Advanced button.

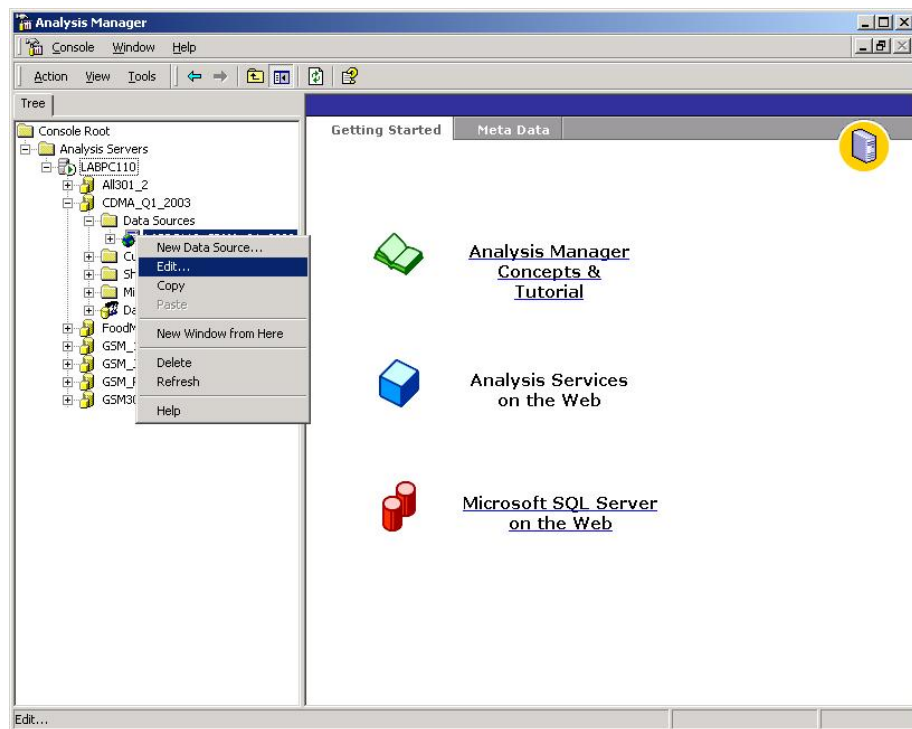


11. Enter connection parameters for the Measurement database: server name, user (= database owner), password, and database name.
12. Click Test Connection. If the test fails, check that the connection parameters are correct.
13. A message "Test connection succeeded" should appear. Click OK to finish the connection test.
14. Click OK to save the new password.
15. In the Geobase properties window, click Cancel.
16. The new Measurement database password for the database owner is now active.
17. Enable all scheduled Geobase processing tasks that were disabled in step **Error! Reference source not found.**

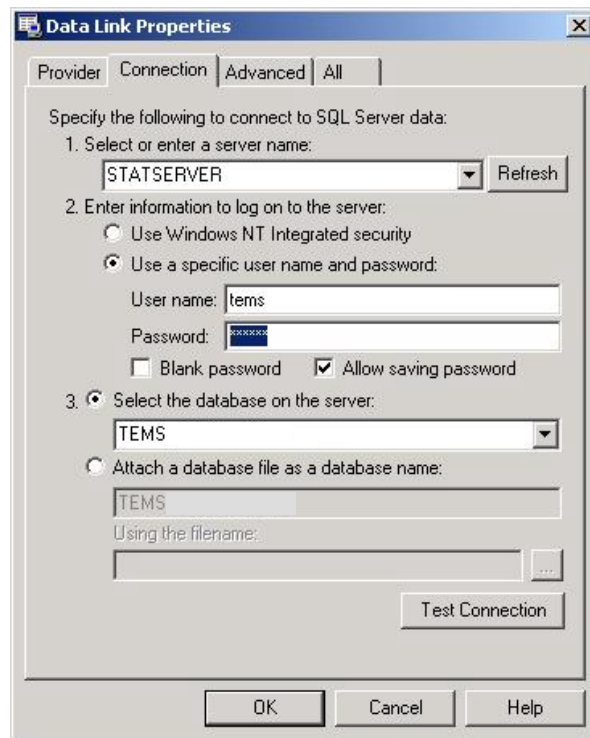
3.5.2. Statistics Database and Geobases

18. Stop the Statistics application and any scheduled Geobase processing tasks that are currently running. This is necessary in order to avoid doing any processing while the passwords are being updated. Disable all scheduled Geobase processing tasks.

19. Stop any other Statistics database processes using the database owner user ("tems").
20. Change the password for the existing Statistics database user (= database owner). See section 4.8, "Changing User Passwords".
21. From the Start menu, open Programs → Microsoft SQL Server → Analysis Services → Analysis Manager.
22. Log on to the OLAP Server as database owner ("tems").
23. Expand as follows: Console Root → Analysis Servers → TA Statistics Server.
24. For the first TEMS Automatic Geobase, expand the Data Sources node.



25. Right-click the data source name and select Edit.



26. Enter the new password.
27. Click Test Connection to verify that the data link parameters will establish a connection to the Statistics database.
28. A message “Test connection succeeded” should appear. Click OK to finish the connection test.
29. Click OK to save the new password.



30. Click OK to acknowledge this security warning.
31. Repeat steps **Error! Reference source not found.** through **Error! Reference source not found.** for the remaining TEMS Automatic Geobases.
32. Enable all scheduled Geobase processing tasks that were disabled in step **Error! Reference source not found.**

4. How to Perform Admin Tasks in Microsoft SQL Server 2005

The default settings for Microsoft SQL Server 2005 should be used. For some installations it might be necessary to adjust the settings. For SQL Server 2005 and Client installation issues, see the TEMS Automatic Database Server Installation Guide.

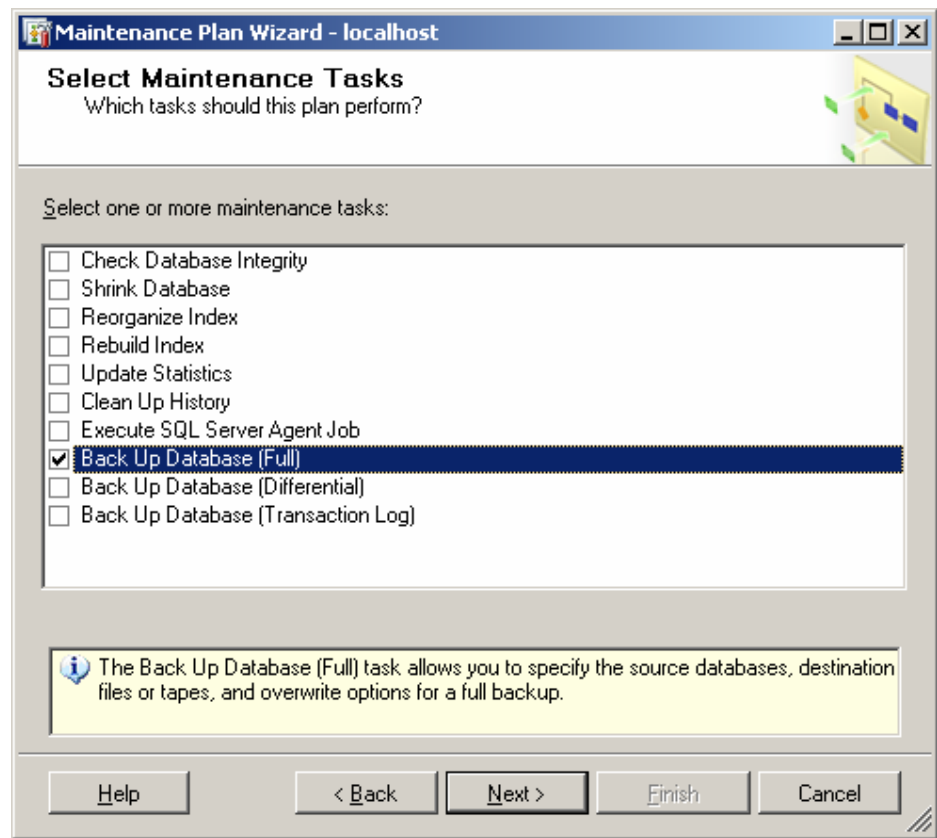
For full details on SQL Server backup and restore procedures, see ref [5].

4.1. Scheduling Database Backups using the Maintenance Plan

The backup of a Microsoft SQL Server database is normally stored on a DAT/DLT tape on the database server. The database server supports a number of backup scenarios, but only the most common ones are described here: full backup to tape device every week and differential backup to disk every day. The latter should be stored on a separate disk, since a disk crash could otherwise destroy both the database and the backup files. If the MSSQLSERVER service is started using a user account instead of the default local system account, it is also possible to store backup files on a network disk.

Note: The Statistics component cannot create or process Geobases while the Statistics database is being backed up.

1. Start SQL Server Management Studio and log in with system administration rights and expand the server where the database you want to create a maintenance plan for is located.
2. Expand Management and right click on Maintenance Plans and select Maintenance Plan Wizard.
3. In the "Welcome to the server maintenance plan wizard" click Next.
4. In the Select Databases dialog, name the maintenance plan, specify server and log in information. Click Next to proceed to the next dialog.



5. Select task Backup Database either Full or Differential and click Next
6. Click Next in the task order dialog.
7. Specify database and where to store the backup in the Backup Database (Full/Differential) Task dialog.
8. Specify the schedule for the maintenance plan and click Next. Please review chapter 1 for schedule settings.
9. Select how and if you want a report from the maintenance plan and click Next.
10. Confirm the settings and click Finish to set up the maintenance plan in the Complete the Wizard dialog.

If the maintenance plan is successfully created close the wizard.

4.2. Scheduling Archiving of Geobases

Geobases should be backed up by two separate methods:

- Full backup of each Geobase, using the archiving function in Microsoft Analysis Services. See ref [4].

- Backup of the entire OLAP Server data disk together with the msmdrep.mdb file of the Microsoft Analysis Services Bin folder. See ref [5].

Both of these methods should be used, with the disk backup as an extra precaution.

Archiving of Geobases is done with the *msmdarch* utility found in the Microsoft Analysis Services Bin folder. For each Geobase, msmdarch creates one or several CAB files on disk. From there the archive files can be moved to the backup media (tape). This can be scheduled by using a SQL Server Agent job.

Note: While existing Geobases are being backed up, it is not possible to create or process Geobases.

Note: The msmdarch utility cannot archive a Geobase where an individual file is larger than 2 GB. If that is the case, the disk backup option has to be relied on.

Note: Before taking a data disk backup, the Microsoft Analysis Service (OLAP) has to be stopped.

4.2.1. Batch File Syntax for msmdarch

This is the batch file syntax for the msmdarch archiving utility. The command takes one argument (%1), the Geobase name:

32bit environment:

```
"C:\Program Files\Microsoft Analysis Services\Bin\
MSMDARCH.EXE" /A [OLAPServerName] "F:\OLAPDataPath\" %1
"E:\OLAPBackupPath\%1.cab" "E:\OLAPBackupPath\%1.log"
```

64bit environment:

```
"C:\Program Files (x86)\Microsoft Analysis Services\Bin\
MSMDARCH.EXE" /A [OLAPServerName] "F:\OLAPDataPath\" %1
"E:\OLAPBackupPath\%1.cab" "E:\OLAPBackupPath\%1.log"
```

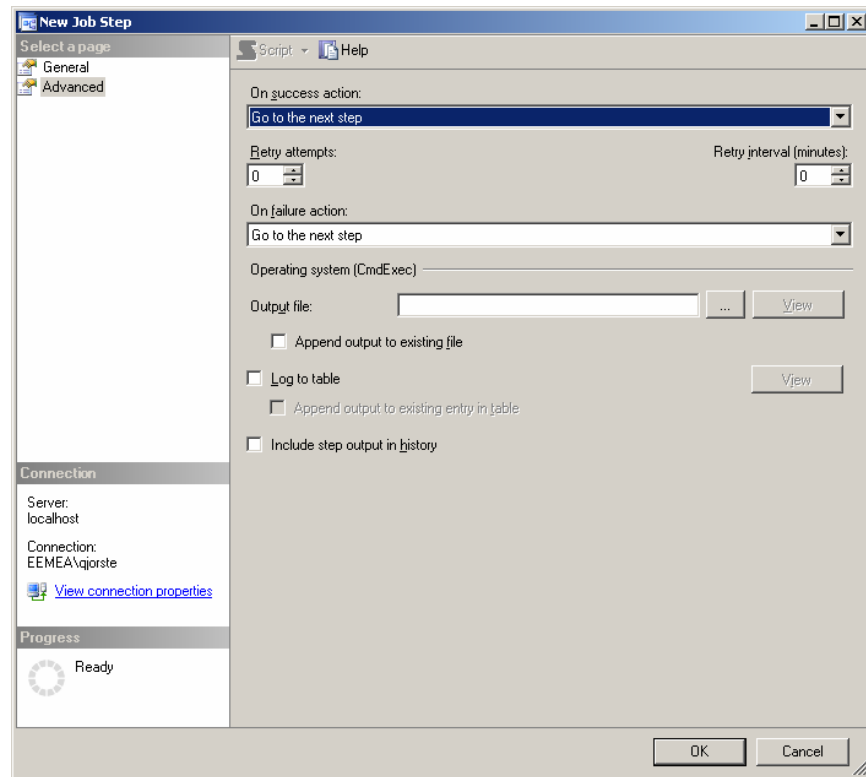
The batch file is named *ArchiveGeobase.bat* in the instructions below, sections **Error! Reference source not found.**–4.2.2

4.2.2. Archiving Geobases using SQL Server Agent Job

All Geobases are archived by a single job, which however must contain one *job step* for each Geobase. See below.

1. Open the SQL Server Management Studio with system administration rights or sa and expand the Statistics database server.
2. Expand SQL Server Agent and the Jobs container
3. Right-click on Jobs and select New Job.
4. Enter name of the job.

5. The job must contain one *step* for each Geobase. Select the Steps page and click New to specify the first step.
6. Under Step name, enter the name of the job step. It is a good idea to reuse the Geobase name. Under Type, select “Operating System(CmdExec)”. In the Command field, type the path to the batch file created in section 4.2.1 followed by the Geobase name.



7. If this is the last Geobase, no more settings are needed; click OK and go to step 10. However, if there are more Geobases, you must first link this step to the next one. Select the Advanced tab.
8. Set both “On success action:” and “On failure action:” to “Goto the next step”. (For the last step you just keep the default settings.) Click OK.
9. To proceed to the next Geobase, click New on the Steps tab and start over with step 6.
10. When you are finished adding Geobases to be backed up, the Steps tab should look something like this. Select the Schedules tab.
11. Enter a name for the schedule. Click new schedule.
12. Select Saturday as the day to perform full backup. Set the time to 04:00 AM. Click OK.
13. Click OK to save the job.

4.3. Restoring a Database using the Maintenance Plan

This section describes how to restore a Microsoft SQL Server database given that the maintenance plan has been implemented according in section 4.1. The full backup may be restored from a tape device or disk. The transaction log backup is restored from a disk.

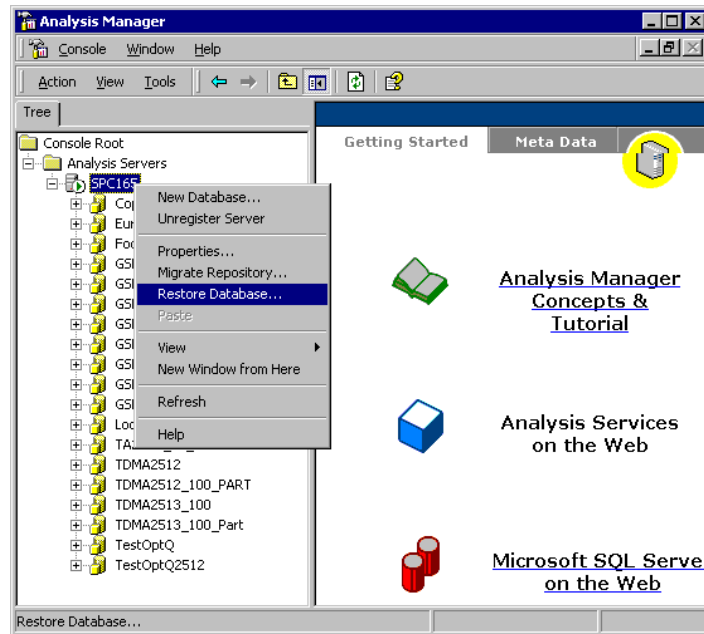
14. Open Microsoft SQL Server Management Studio and expand the server where the database you want to restore is located.
15. Right-click on the database and select **Tasks -> Restore -> Database** from the popup menu.
16. Select the last backup set, including both the full backup and all subsequent backups and click OK.
17. The Restore operation starts.
18. A success message should appear click OK.
19. The database restore operation is now finished.

4.4. Restoring Geobases

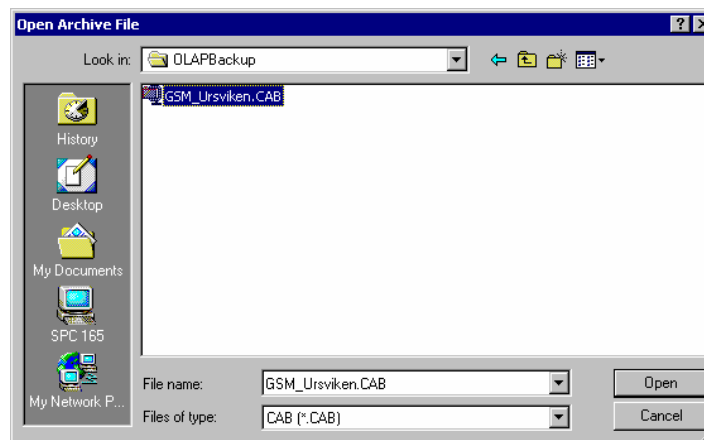
Geobases must be restored one by one manually.

An OLAP Server Geobase is normally restored from a disk file (*.cab) after being uploaded from the backup media, i.e. tape. If for some reason the CAB files cannot be used for recovery, the Geobases have to be restored from the OLAP data file structure backup (see section 4.2). In the latter case, the OLAP Server has to be stopped and the file structure together with the msmdrep.mdb file must be used to overwrite the complete OLAP data file structure.

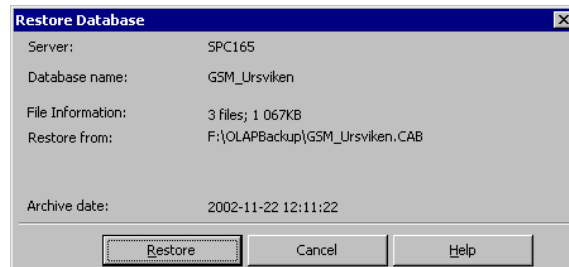
1. Open the Analysis Manager and expand the server where the Geobase you want to restore is located.



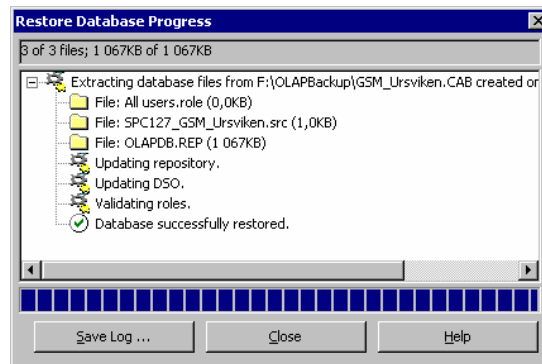
2. Right-click the server name and select Restore Database from the popup menu.



3. Select the first CAB file of the Geobase to restore. Click Open. If the Geobase already exists, select Overwrite.



4. Click Restore.



5. When the restore operation has completed, click Close.

The Geobase is now restored.

4.5. Backing Up a Database Manually

This section shows how to make a full backup of a database manually.

1. Open SQL Server Management Studio and expand the server where the database you want to back up is located.
2. Right-click the database and select **Tasks -> Back Up** from the popup menus.
3. Select backup type Full, Differential or Transaction Log.
4. Select destination and click OK to start the backup.
5. When the backup is complete, a success message appears. Click OK.
6. The backup operation is now finished.

4.6. Restoring a Database Manually

This section shows how to perform a manual restore of a database.

1. Open Microsoft SQL Server Management Studio and expand the server where the database you want to restore is located.
2. Right-click on the database and select **Tasks -> Restore -> Database** from the popup menu.
3. Select the last backup set, including both the full backup and all subsequent backups and click OK.
4. The Restore operation starts.
5. A success message should appear click OK.

6. The database restore operation is now finished.

4.7. Adding a User

Here is how to add a new user in the TEMS Automatic system.

1. Launch SQL Server Management Studio and log on as database owner *tems*.
2. Expand the server item as follows: **Security** -> **Logins**. Right-click and select New Login.
3. Select SQL Server login.
4. Specify name and password for the new user.
5. Select the User Mapping page.
6. Select the checkbox next to a database and a role to map the login to that role in the selected database.
7. Review the tools and database roles tables in figure 3 and figure 4 to assign the correct role to the login.

Tool	User Profile
Presentation	Measurement and Statistics
Operator Console	Measurement
Call Generator	Measurement
ComServer	Measurement
Logfile Handler (TLH)	Measurement
Service Monitor	Measurement

Figure 3 Table 4:1

Tool	User Profile	Database Roles
Presentation	Normal user	TEMS_Presentation_User
Presentation	Site Import	TEMS_Automatic_Administrator , TEMS_Presentation_User, TEMS_Automatic_TLH
Operator Console	Administrator user	TEMS_Automatic_Administrator
Operator Console	Normal user	TEMS_Automatic_Operator
Operator Console	Read-only user	TEMS_Automatic_Monitor
Call Generator	Application	TEMS_Automatic_CallGenerator
ComServer	Application	TEMS_Automatic_ComServer
Logfile Handler (TLH)	Application	TEMS_Automatic_TLH
Service Monitor	Application	TEMS_Automatic_Administrator

Figure 4 Table 4:2

8. Click OK to create the new login.

4.8. Changing User Passwords

The *Call Generator*, *ComServer*, *Logfile Handler*, and *Service Monitor* needs to be in a stopped state and logged off from the database before the passwords for their database users can be changed.

For the Call Generator and ComServer the password change can be done from the application user interface.

For the Logfile Handler and Service Monitor SQL Server Management Studio must be used to change the password. Follow the steps below.

1. Launch SQL Server Management Studio and log on as the user to password for or as database owner *tems*.
2. Expand the server item as follows: *Security* -> *Logins* and right click on the login and select Properties.
3. Enter the new password and confirm it.
4. Click OK.

4.9. Altering User Rights

Note: Assigning multiple roles to one user is *not* recommended for users representing services. Such an assignment is applicable only for normal users.

1. Launch Service Monitor SQL Server Management Studio and log on as database owner ("tems").
2. Expand the server item as follows: *Security* → *Logins*. Right-click the user and select Properties. Select the Database Access tab.
3. Select the databases which the new user should be able to access. Different users need access to different databases as described in the table below.

Tool	User Profile
Presentation	Measurement and Statistics
Operator Console	Measurement
Call Generator	Measurement
ComServer	Measurement
Logfile Handler (TLH)	Measurement
Service Monitor	Measurement

Figure 5 Table 4:3

4. For each database selected, add the roles the user should have. Different users are given different access rights to the databases. The access rights are controlled with the help of user roles as described in the table below. The

DBClean tool and the Statistics tool are operated by the database owner ("tems").

Tool	User Profile	Database Roles
Presentation	Normal user	TEMS_Presentation_User
Presentation	Site Import	TEMS_Automatic_Administrator , TEMS_Presentation_User, TEMS_Automatic_TLH
Operator Console	Administrator user	TEMS_Automatic_Administrator
Operator Console	Normal user	TEMS_Automatic_Operator
Operator Console	Read-only user	TEMS_Automatic_Monitor
Call Generator	Application	TEMS_Automatic_CallGenerator
ComServer	Application	TEMS_Automatic_ComServer
Logfile Handler (TLH)	Application	TEMS_Automatic_TLH
Service Monitor	Application	TEMS_Automatic_Administrator

Figure 6 Table 4:4

5. Click OK. Confirm the password if requested.

4.10. Connecting Database Users and Logins

If the users in a database (such as Measurement or Statistics) have no connection to the login information stored in the master database (happens when attaching a database, moved from another database server), there is a need to connect the two. The system-stored procedure `sp_change_users_login` can be run in order to find database users with no logins and makes the connection.

To find the missing connections, run the command below as database owner ("tems") from SQL Query Analyzer:

```
EXEC sp_change_users_login 'Report'
```

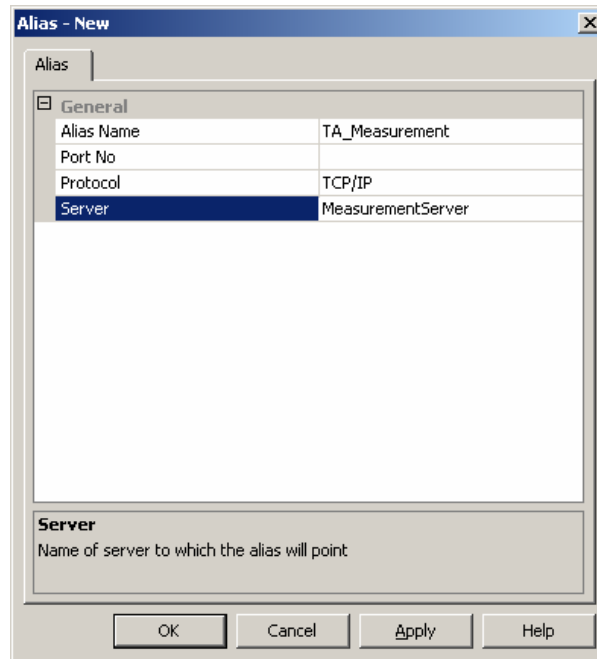
For each user found, run the following command with the user as second argument and the desired login as third argument (both = ServiceMonitor in this example):

```
EXEC sp_change_users_login 'Update_One', 'ServiceMonitor',  
'ServiceMonitor'
```

4.11. Creating a Linked Server

- A. On the SQL Server running the Statistics database, start the **SQL Server Configuration Manager**.
- B. Expand *SQL Native Client Configuration* and right click on *Aliases* and select **New Alias**
- C. In the *Alias Name* and *Server* fields, type the name of the Measurement database server. Select the TCP/IP network libraries and click OK.

- D. Right click and select **New Alias** again.
- E. In the *Alias* field, enter the name of the Measurement database server with *TA_* prefixed to it. In the *Server* field, enter the Measurement database server name as-is. Select the TCP/IP network libraries and click OK.



- F. Start *SQL Server Management Studio* and log in as *sa*:
- G. Expand the SQL Server and *Server Objects* container and right click on *Linked Servers* and select **New Linked Server...**
- H. In the *Linked Server* field, type *TA_<name of Measurement database server>*. Under *Server Type*, select **SQL Server** and click on the *Security* page.
- I. Select *Be made using this security context* and type in the user tems as *Remote login* and the accompanying password. Then click the *Server Options* page.
- J. Enable the options for **Data Access**, **RPC**, **RPC Out**, and **Use Remote Collation**. Then click OK to create the linked server.

4.12. Changing the Remote Query Timeout Value

- A. In *SQL Server Management Studio*, make sure you are logged in as *sa* on the SQL Server running the Measurement database.
- B. Right-click on the server select *Properties* and then select the *Connections* page.

- C. In the *Remote server connections* section set *Query time-out* to **0**.
- D. Restart the Measurement database server.
- E. If the Statistics database is on the same SQL server as the Measurement database, you are done. If the Statistics database resides on a different SQL server, you need to repeat the actions above for that server.

4.13. Administrating Logfile Storage Databases

A Logfile Storage database stores logfiles and information about logfiles (i.e. Logfile name, bounding box, time interval). The storage structure consist of a number of table sets holding logfiles. A table set is the two tables tLogfileStorage_YYYYMM and tLogfileBlob_YYYYMM, where YYYY represents the year and MM the month (01-12). Each table set stores logfiles generated from one and only one month. The Logfile Storage database(s) belongs to one and only one Measurement database. The content of the Logfile Storage database(s) are accessed through two views (vLogfileStorage and vLogfileBlob) in the Measurement database. Logfile Storage databases as well as the logfile storage tables sets can be added to a TEMS Automatic system during the system's life span. The DBClean application is used to remove tables (and as a consequence the stored logfiles) from the system.

It is recommended that a Logfile Storage database contain a limited set of tables covering a limited time range (3–6 months).

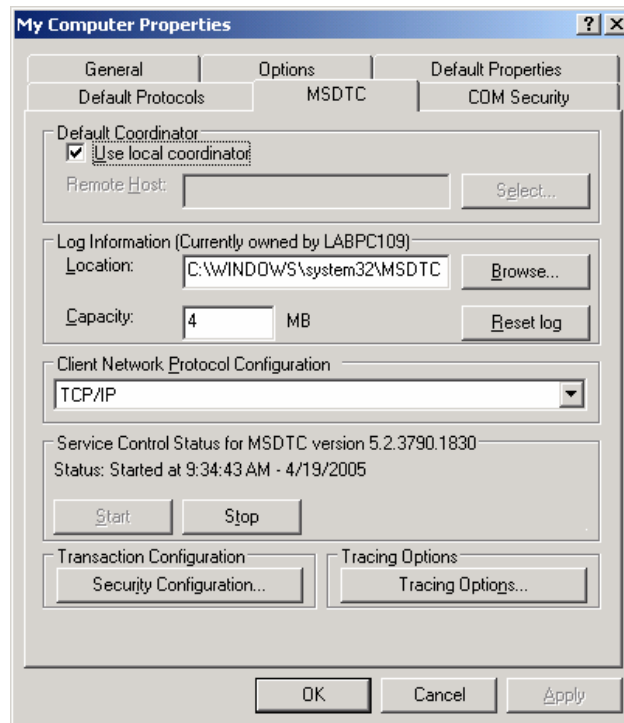
A Logfile Storage database can reside on the same database server as the Measurement database, or it can reside on another server. All Logfile Storage databases must have the same dbo (and dbo password) as the Measurement database. All users of the Measurement database have to be replicated to all Logfile Storage databases.

If a Logfile Storage database resides on another database server than the Measurement database, the Distributed Transaction Coordinator (DTC) has to be started on both servers. The Measurement database has to be installed on SQL Server 2005 Enterprise Edition. If the SQL Server is running on a Windows 2003 operating system the default settings for the DTC has to be altered, see below.

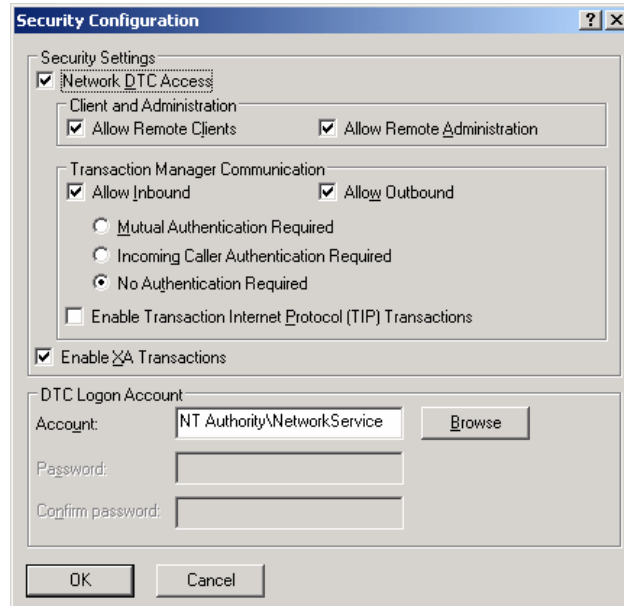
All Logfile Handler instances have to be stopped when performing any of the Logfile Storage administration tasks listed below.

4.13.1. Enabling DTC on Windows 2003 (SP1)

1. Start the Component Services by selecting Start → Programs → Administrative Tools → Component Services on the SQL Server running Windows 2003 (SP1).
2. Open Component Services/Computers.
3. Right click My Computer and select Properties.
4. Select tab MSDTC.



5. Select Security Configuration.



6. Select Network DTC Access.

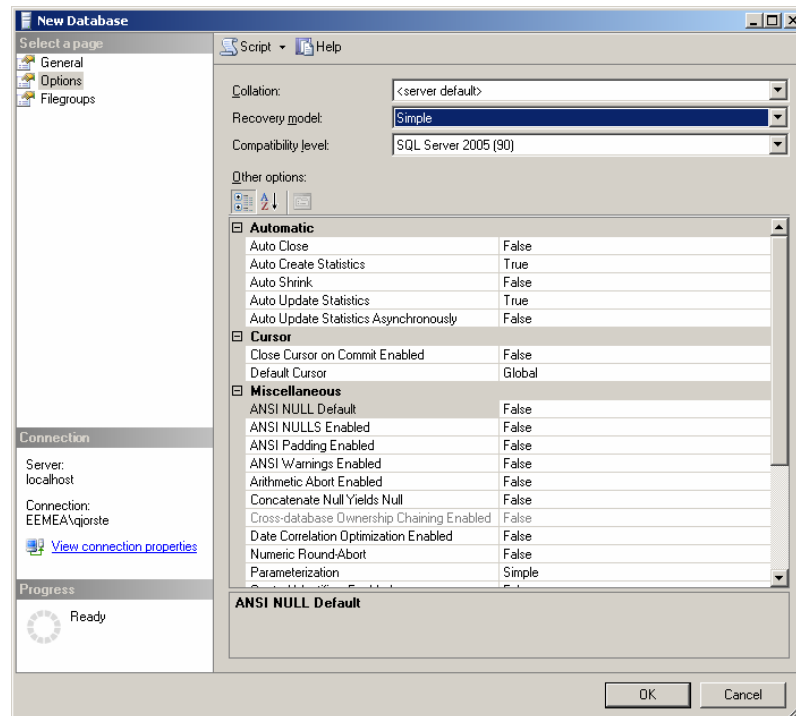
7. Make sure that DTC Logon Account is set to NT Authority\NetworkService.

8. Click OK.

9. Click OK.
10. Close the Component Services application.

4.13.2. Creating a Logfile Storage Database

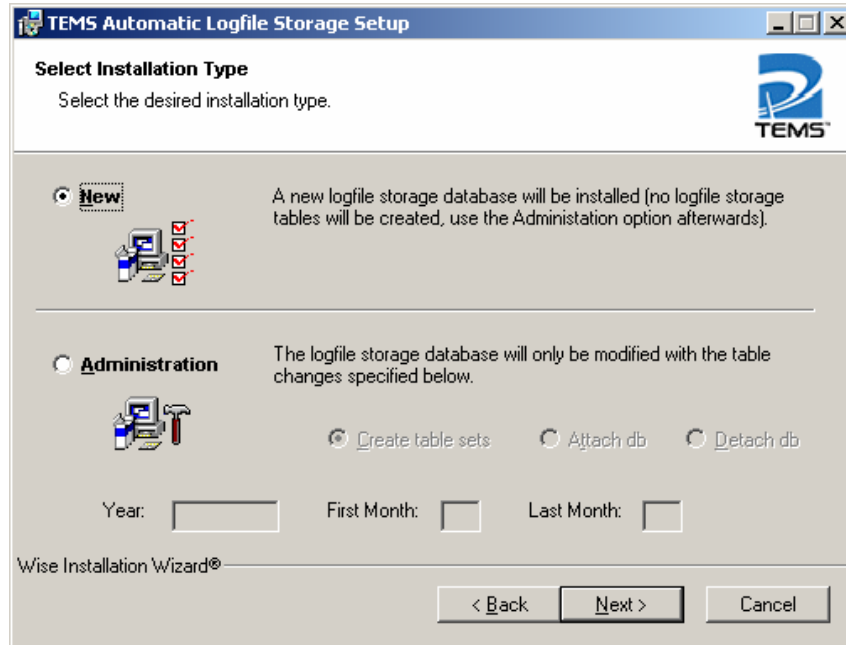
1. Open the Microsoft SQL-Server Management Studio.
2. Log in as "tems". (Right-click the server icon, select "Edit SQL Server Registration properties", select "Use SQL Server authentication", and enter the login name and password of the tems user. Then click OK).
3. Right-click the Databases item and select New Database.
4. Specify the name of the database to be created. Be sure to avoid using names listed in the *Reserved Keywords (T-SQL)* document in SQL Server Books Online. Do not use the space character, and do not use more than 20 characters.
5. Change the initial value for the database file to **512**.
6. Edit the *Autogrowth* and select **In megabytes** and change the value to 512.
7. Specify the location for the datafile and the transaction log file.
8. Select the default settings for the transaction log file.



9. Click the Options page and select the simple recovery model.

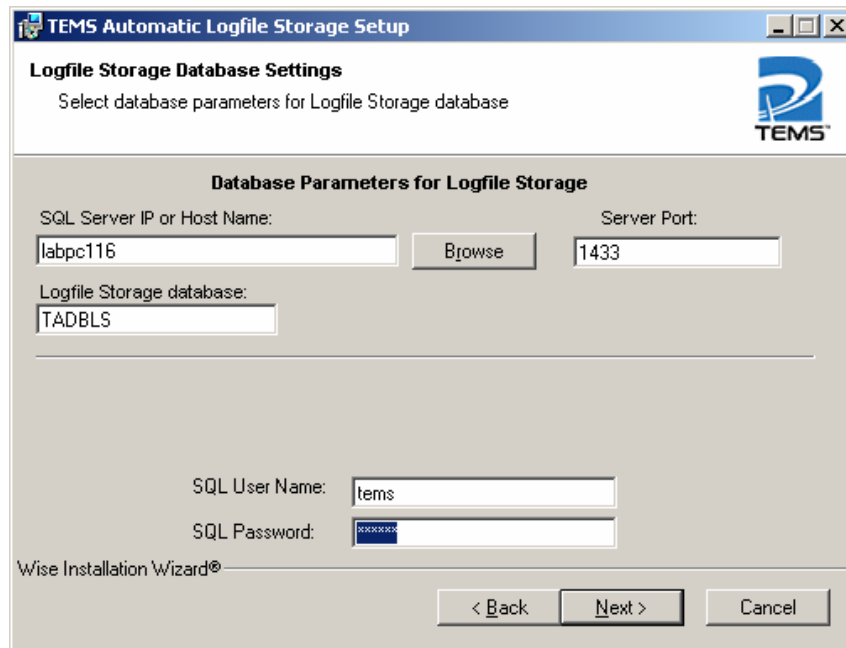
10. Click OK.

11. Run the `Logfile Storage.exe` file located on the logfile storage installation disk. The welcome screen appears click Next to continue.



The screenshot shows the 'Select Installation Type' window of the 'TEMS Automatic Logfile Storage Setup' wizard. The window has a title bar with the application name and standard window controls. Below the title bar, the text 'Select the desired installation type.' is displayed. There are two radio button options: 'New' (selected) and 'Administration'. The 'New' option includes a description: 'A new logfile storage database will be installed (no logfile storage tables will be created, use the Administration option afterwards).' and an icon of a computer with a plus sign. The 'Administration' option includes a description: 'The logfile storage database will only be modified with the table changes specified below.' and an icon of a computer with a hammer. Below the 'Administration' option, there are three radio buttons: 'Create table sets' (selected), 'Attach db', and 'Detach db'. At the bottom, there are three text boxes labeled 'Year:', 'First Month:', and 'Last Month:'. The 'Wise Installation Wizard' logo is in the bottom left corner. At the bottom right, there are three buttons: '< Back', 'Next >', and 'Cancel'.

12. Select New. Click Next.

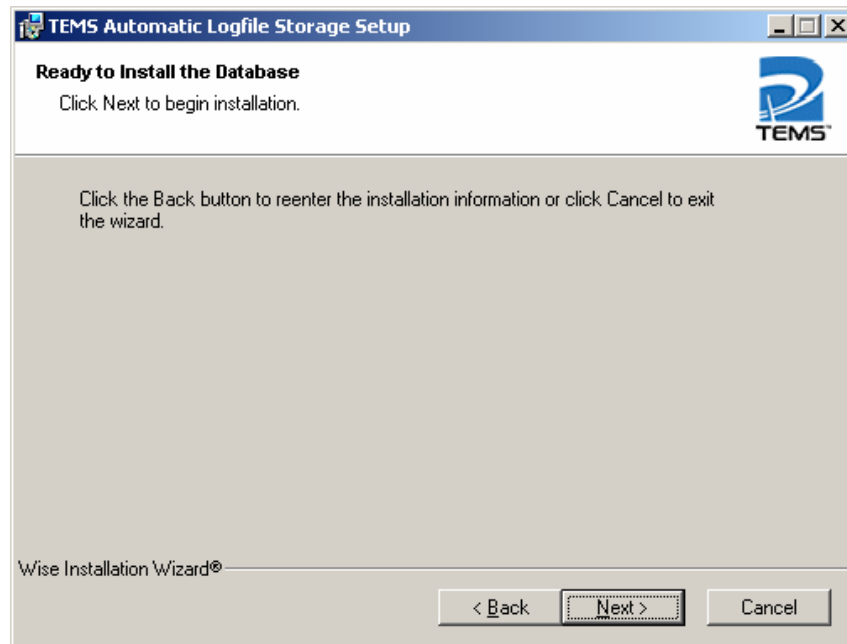


The screenshot shows the 'Logfile Storage Database Settings' window of the 'TEMS Automatic Logfile Storage Setup' wizard. The window has a title bar with the application name and standard window controls. Below the title bar, the text 'Select database parameters for Logfile Storage database' is displayed. The main section is titled 'Database Parameters for Logfile Storage'. It contains several fields: 'SQL Server IP or Host Name:' with the value 'labpc116' and a 'Browse' button; 'Server Port:' with the value '1433'; 'Logfile Storage database:' with the value 'TADBLS'; 'SQL User Name:' with the value 'tems'; and 'SQL Password:' with a masked password 'xxxxxx'. The 'Wise Installation Wizard' logo is in the bottom left corner. At the bottom right, there are three buttons: '< Back', 'Next >', and 'Cancel'.

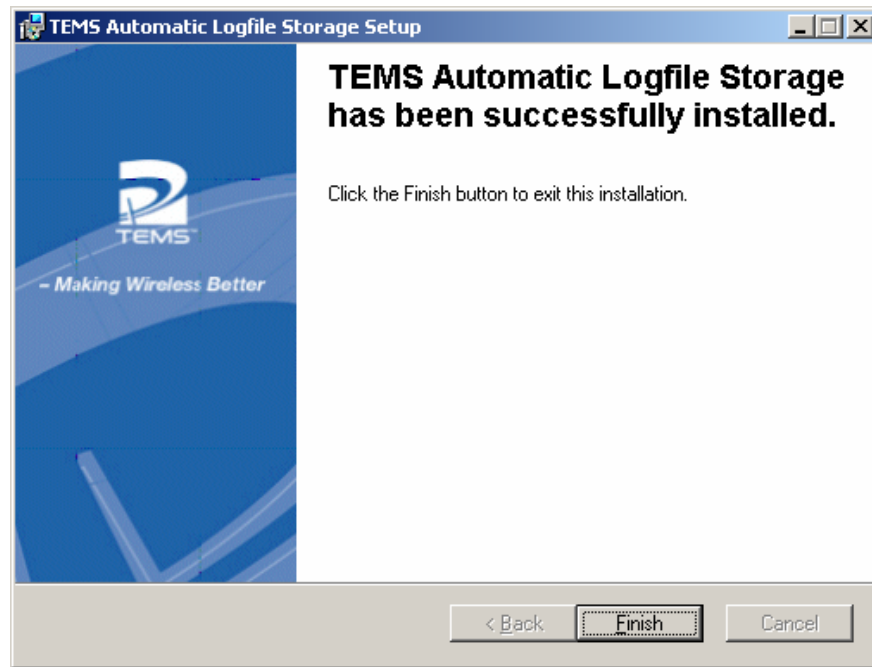
13. Fill out all fields.

- **SQL Server IP:** This is the name or IP address of the database server machine.
- **Server Port:** By default set to 1433, the port most commonly used for MS SQL Server 2000. However, please verify this by checking the database server installation.
- **Database:** The name of the Logfile Storage database that was created before the installation started.
- **User:** The user who is database owner.
- **Pwd:** Password for the database owner user.

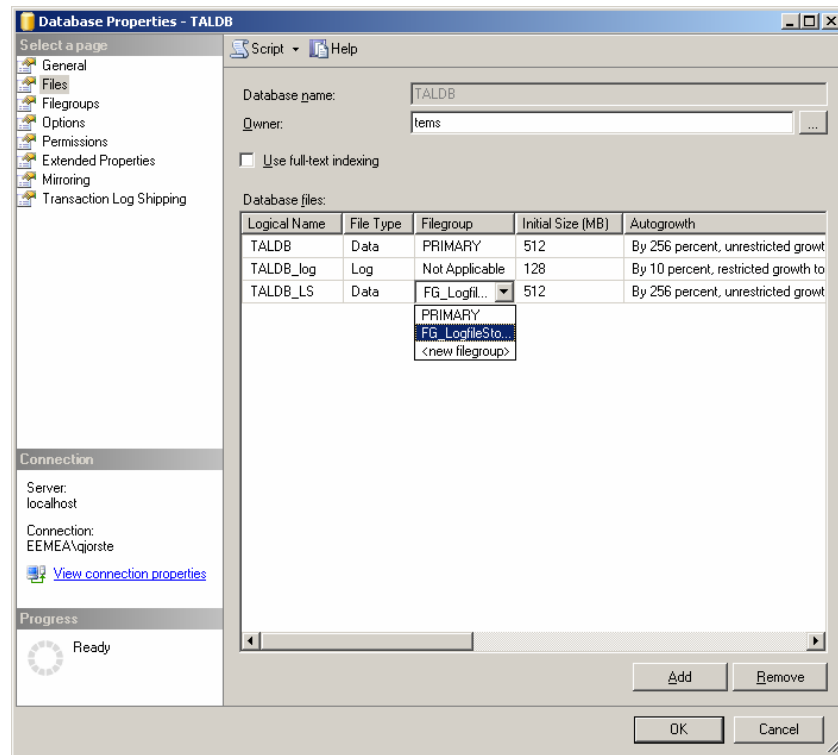
14. Click Next.



15. Click Next.



16. Click Finish.
17. Right-click the Logfile Storage database in the SQL Server Management studio and select Properties (to create a physical database file for the Logfile Storage file group).
18. Click the Files tab



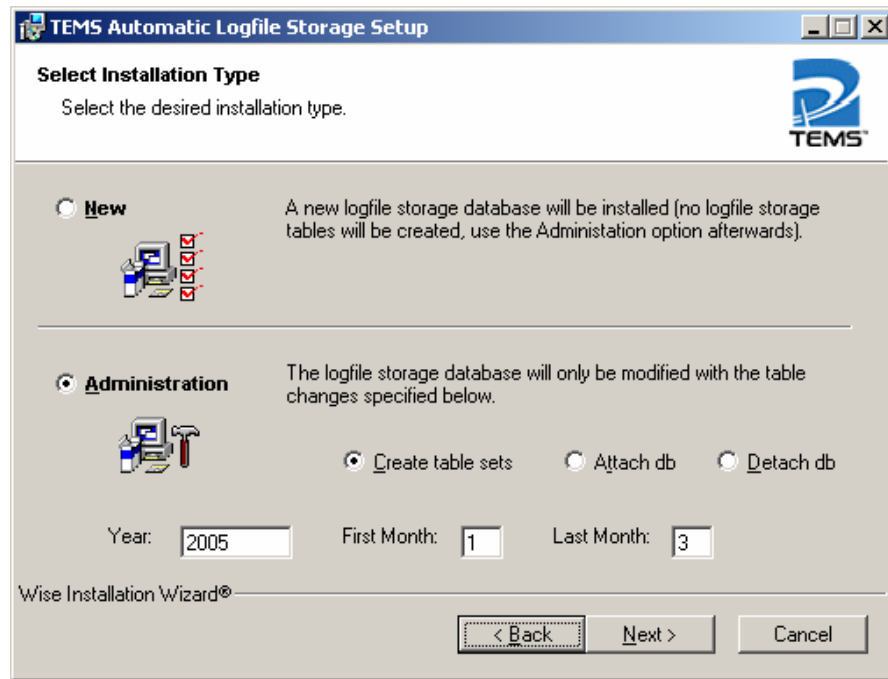
19. Create a new physical database file for the Logfile Storage file group. Change the value in the “Space allocated (MB)” column to the value recommended in the TEMS Automatic Dimensioning Guide. (This is a support document; refer to TEMS Support for guidance.)

Example: The system has 24 GSM MTUs running for 8 hours a day, 25 days a month, and using a work order with 120 s voice and 15 s idle mode measurements. Logfiles in logfile storage (blobs) are saved for 1 month before being backed up and removed from the database. In this case you should set “Space allocated” to at least 10,000 MB.

20. Change the value in the “Filegroup” column to FG_LogfileStorage.
21. Click OK (proceed with creating logfile storage tables).

4.13.3. Creating Logfile Storage Tables

1. Run the `Set up . exe` file located on the logfile storage installation disk. The welcome screen appears.



2. Select Administration.

3. Fill out all fields.

- **Year:** This is year of the logfile to be stored.
- **First Month:** The first month in the table set.
- **Last Month:** The last month in the table.

4. Click Next (pressing Tab to leave the Last Month field will enable the next button).

TEMS Automatic Logfile Storage Setup

Measurement Database Settings
Select database parameters for Measurement database

Database Parameters for Measurement

SQL Server IP or Host Name: Server Port:

Measurement database:

SQL User Name:

SQL Password:

Wise Installation Wizard®

< Back Next > Cancel

6. Fill out all fields.

- **SQL Server IP:** This is the name or IP address of the *Measurement* database server machine.
- **Server Port:** By default set to 1433, the port most commonly used for MS SQL Server 2000. However, please verify this by checking the database server installation.
- **Database:** The name of the *Measurement* database.
- **User:** The user who is database owner.
- **Pwd:** Password for the database owner user.

7. Click Next.

TEMS Automatic Logfile Storage Setup

Logfile Storage Database Settings
Select database parameters for Logfile Storage database

Database Parameters for Logfile Storage

SQL Server IP or Host Name: Server Port:

Logfile Storage database:

SQL User Name:

SQL Password:

Wise Installation Wizard®

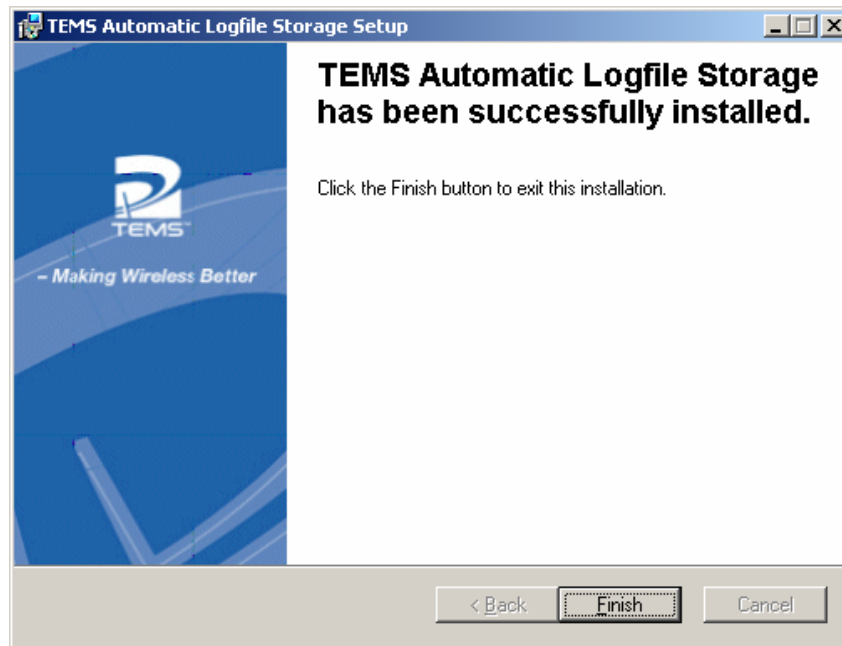
< Back Next > Cancel

8. Fill out all fields.

- **SQL Server IP:** This is the name or IP address of the *Logfile Storage* database server machine.
- **Server Port:** By default set to 1433, the port most commonly used for MS SQL Server 2000. However, please verify this by checking the database server installation.
- **Database:** The name of the *Logfile Storage* database.
- **User:** The user who is database owner.
- **Pwd:** Password for the database owner user.

9. Click Next.

10. Click Next.

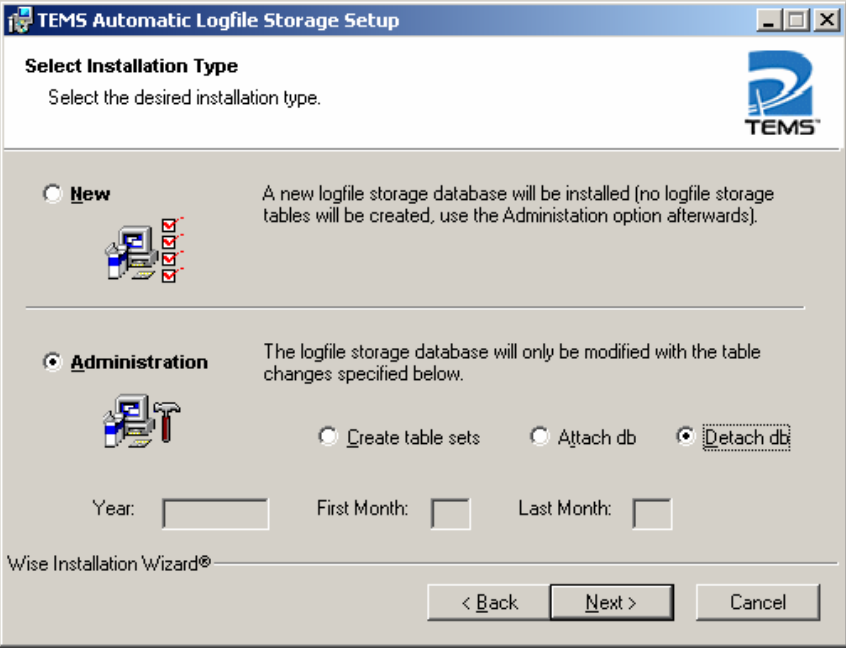


11. Click Finish.

4.13.4. Attaching/Detaching a Logfile Storage Database

After detaching a Logfile Storage database, the Logfile Storage database will not be used any more by the TEMS Automatic system; it is no longer accessible from the Measurement database. The Logfile Storage database still exists on the database server and can be attached again at a later time. This feature is supplied so that a Logfile Storage database with old logfiles (not currently used) can be moved to a backup media to preserve disk space.

1. When attaching a Logfile Storage database, check that the Logfile Storage database still exists on the database server and that the dbo is the same as the dbo for the Measurement database.
2. Run the `Setup.exe` file located on the logfile storage installation disk. The welcome screen appears click Next to continue.



TEMS Automatic Logfile Storage Setup

Select the desired installation type.

☐ **New** A new logfile storage database will be installed (no logfile storage tables will be created, use the Administration option afterwards).

☒ **Administration** The logfile storage database will only be modified with the table changes specified below.

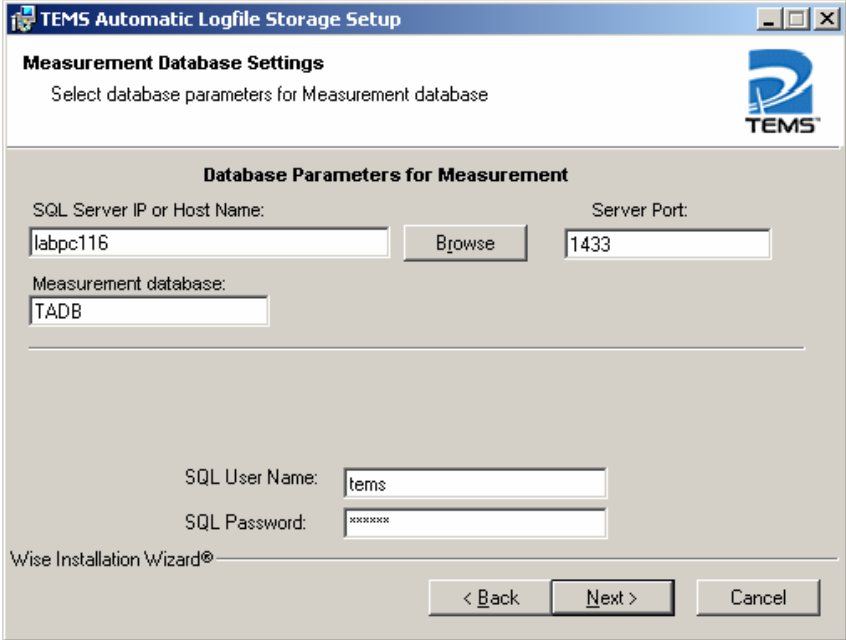
☐ Create table sets ☐ Attach db ☒ Detach db

Year: First Month: Last Month:

Wise Installation Wizard®

< Back Next > Cancel

3. Select Administration.
4. Select Detach db or Attach db.
5. Click Next.



TEMS Automatic Logfile Storage Setup

Select database parameters for Measurement database

Database Parameters for Measurement

SQL Server IP or Host Name: Browse Server Port:

Measurement database:

SQL User Name:

SQL Password:

Wise Installation Wizard®

< Back Next > Cancel

6. Fill out all fields.

- **SQL Server IP:** This is the name or IP address of the *Measurement* database server machine.
- **Server Port:** By default set to 1433, the port most commonly used for MS SQL Server 2000. However, please verify this by checking the database server installation.
- **Database:** The name of the *Measurement* database.
- **User:** The user who is database owner.
- **Pwd:** Password for the database owner user.

7. Click Next.

The screenshot shows a Windows-style dialog box titled "TEMS Automatic Logfile Storage Setup". Inside, the section "Logfile Storage Database Settings" is active, with the instruction "Select database parameters for Logfile Storage database". The "Database Parameters for Logfile Storage" section contains the following fields and controls:

- SQL Server IP or Host Name:** A text box containing "labpc116" and a "Browse" button.
- Server Port:** A text box containing "1433".
- Logfile Storage database:** A text box containing "TADBLS".
- SQL User Name:** A text box containing "tems".
- SQL Password:** A text box with masked characters (xxxxxx).

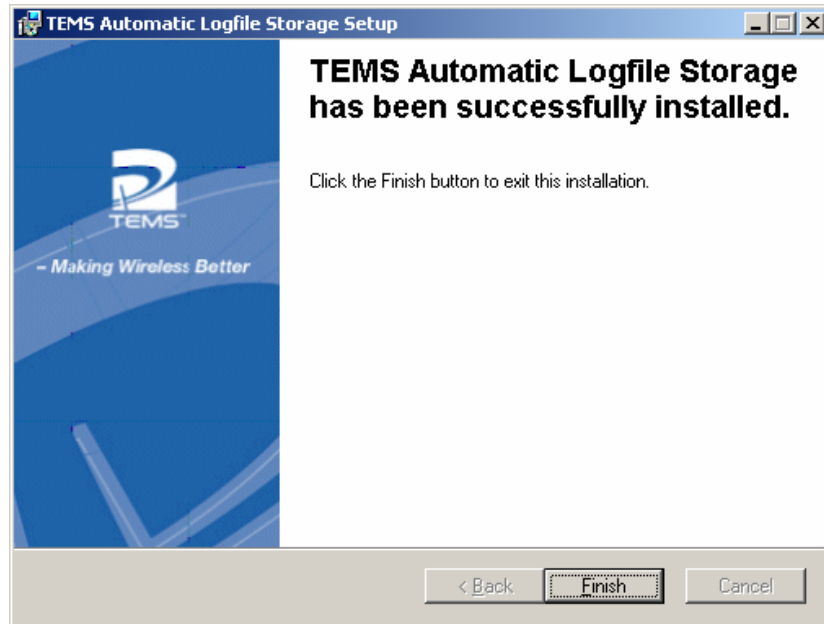
At the bottom, the text "Wise Installation Wizard®" is visible. The navigation buttons are "< Back", "Next >" (which is highlighted with a blue border), and "Cancel". The TEMS logo is in the top right corner.

8. Fill out all fields.

- **SQL Server IP:** This is the name or IP address of the *Logfile Storage* database server machine.
- **Server Port:** By default set to 1433, the port most commonly used for MS SQL Server 2000. However, please verify this by checking the database server installation.
- **Database:** The name of the *Logfile Storage* database.
- **User:** The user who is database owner.
- **Pwd:** Password for the database owner user.

9. Click Next.

10. Click Next.



11. Click Finish. The detached Logfile Storage database can now be removed from the database server.

4.14. What to Do If the Database File Does Not Grow

This section describes troubleshooting for the situation where the automatic growth of the database files is not working.

1. Start SQL Server Management Studio and create a new query.
 - a. In the new query window write `EXEC sp_helpdb '<dbname>'`
 - b. Press *F5* to execute, you will now get information about your database and database files in the result window
 - c. Look in the *size*, *maxsize* and *growth* columns for your database files.
 - d. If *growth* is 0 KB it has been disabled.
 - e. Also check so that the file has not reached the *maxsize*.
2. Create a new query and execute the following statement, `DBCC SQLPERF ( LOGSPACE )`
 - a. This will show the size of the logfile and how much space that is used by logged transactions.

- b. Confirm that the space used is not 100% or close to 100%.
3. Create a new query and execute the following statement, EXEC
sp_spaceused
 - a. Check if there is any unused space for the the database.
4. If these things are OK, then there may be some other problem with this specific file, for example a problem writing to the drive. To remedy this situation, we need to create a second data file that will start adding data to the database. It will be part of the same database, and the data collected on the first file will be usable.
 - a. First stop all TEMS Automatic services so that no new logfiles are coming in.
 - b. In the *Database Properties* dialog on the *Files* page, disable the *Autogrowth* option for the first database file.
 - c. Add a new database file by clicking on the *Add* button. In the *Initial size* column, enter 512 or 1024 MB. Enable *Autogrowth* specify **In megabytes** and enter a small value such as **50 MB**. This is just to quickly find out whether the workaround is going to work. Under Maximum file size, select **Unrestricted file growth**. Click OK.
 - d. Now restart the TLH and other services. See whether the database file starts growing automatically in the intended way.

5. Recurring Tasks

In this chapter are listed administration tasks that should be repeated on a daily, weekly, or monthly basis.

System backup is described in detail in chapters 2 and 4 and is only briefly mentioned here.

5.1. Daily Tasks

5.1.1. General

Check the Windows Event Viewer for warnings and errors produced by the TEMS Automatic system. This should be done on all machines, i.e. those where the ComServer, Call Generator, Logfile Handler, database, FTP server, and RAS server are located. Note that some of these are likely to be on the same physical machine.

In Presentation, check the Geobase properties in the statistics configuration window and verify that it has been processed as expected.

5.1.2. Operator Console

5.1.2.1. Test Units (TUs)

Check for TU groups tagged red (alarm) or yellow (warning). Expand any such groups and look for red or yellow TUs. Access the Properties dialog for each of these TUs and click the Event History tab to find the cause of the alarm or warning. Take appropriate action (as suggested by the Operator Console online help). When ready, acknowledge the warning or alarm.

MTU name	Alarm	State	MTU type	Serial number	Software revision	Hardware
23112368220			TDMA	23112368220	TDMA R2.5.0.5 ...	TDMA PC
23112374017			TDMA	23112374017	TDMA R2.5.0.5 ...	TDMA PC
23112374415			TDMA	23112374415	TDMA R2.5.0.5 ...	TDMA PC

5.1.2.2. Servers

Check for servers tagged red. Investigate the cause and take appropriate action (as suggested by the Operator Console online help).

Servers - Server information			
Servers	Server name	Type	Alarm Latest access
Communication Server	EF8HP01C6G3NK8I	Communication Server	2002-12-13 11:22:58
Logfile Handler	spc121_3	Logfile Handler	2002-12-13 11:23:32
	spc071	Logfile Handler	2002-12-13 11:21:50
	spc071_1	Logfile Handler	2002-12-13 11:22:22
	spc121	Logfile Handler	2002-12-13 11:24:09
	spc121_1	Logfile Handler	2002-12-13 11:23:26
	spc121_2	Logfile Handler	2002-12-13 11:24:04
Call Generator	SPC175	Call Generator	2002-12-13 11:18:40

5.1.3. ComServer

Verify that files are moved from the In folder on the FTP server and that the ComServer icon in the system tray is blinking at regular intervals (by default: every 30 s).



5.1.4. Logfile Handler

Check that logfiles are processed by checking that files are moved from the Logfiles folder to the Converted folder, so that the number of files in the Converted folder is increasing.

Also check the number of files in the Garbage folder. If the number of files is increasing rapidly, this might indicate disturbances in the system.

5.1.5. Call Generator

Use the Call Generator Channel Information view in the Operator Console to check that the channels are working. To display this view, locate the Call Generator in the Servers tree view and double-click it. See below.

Servers - Call Generator channel information								
Servers	Channel	Alarm	State	Hook	MTU phone number	PSQM UL	Mean usage	Start time
Communication Serve	B1C1		Idle	On	-	0 %	2002-12-09 16:58:50	
Logfile Handler	B1C2		Idle	On	-	0 %	2002-12-02 11:11:46	
Call Generator	B1C3		Idle	On	-	0 %	2002-12-02 12:25:02	
	B1C4		Idle	On	-	0 %	2002-12-02 11:11:51	
	B2C1		Idle	On	-	0 %	2002-12-03 12:30:44	
	B2C2		Idle	On	-	0 %	2002-12-09 16:58:31	

If there are any alarms, select the phone line and click the button with the red bell to view all alarms.

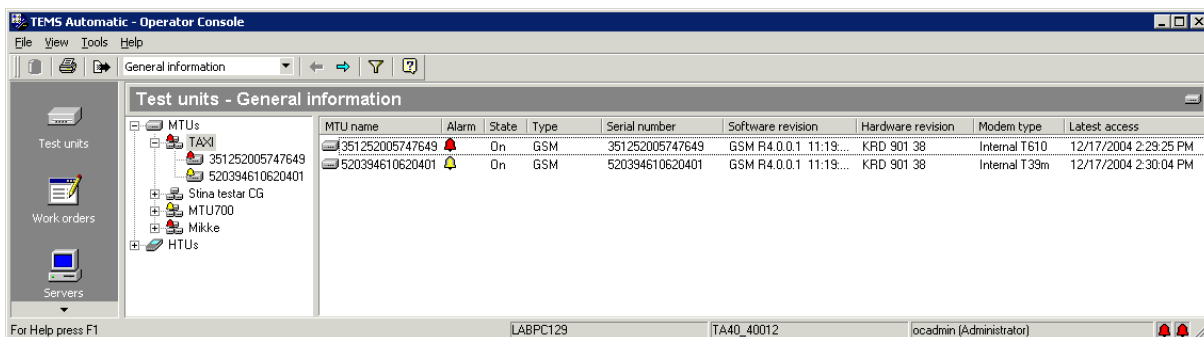
5.1.6. Backup

Verify that the daily backup of the database transaction log has been made as specified in the maintenance plan. See sections 2.1 and 4.1.

5.2. Weekly Tasks

5.2.1. Operator Console

Check that the TUs are measuring by accessing the information view “Test units – General information” and inspecting the Latest access column. If an MTU has not made contact for a long time, check that the car carrying the MTU is in service. If a TU has not reported any measurements at all, make sure that the assigned work order is valid and has been accepted by the TU.



5.2.2. Measurement Database

If the Measurement database server is running low on disk space, remove old data using Database Clean, AutoDBClean or add more hard disk capacity to the server.

5.2.3. Statistics Database

If the Statistics database server is running low on disk space, remove some partitions or add more hard disk capacity to the server.

5.2.4. RAS Server

Verify that all incoming phone lines are working. The RAS server should answer an incoming call.

Check the Windows Event Log for invalid logins by TUs and/or improperly working modems.

5.2.5. Logfile Handler

If necessary, compress processed files in the Converted folder, and move files out of this folder if disk space is lacking.

(In systems of moderate size it might be enough to do this once a month.)

5.2.6. Backup

Verify that the weekly full backup of databases, Geobases and folders has been made as specified in the maintenance plan. See sections 2.1 and 4.1.

5.3. Monthly Tasks

5.3.1. Logfile Handler

Compress and, if space is lacking, move

- processed files from the Converted folder. (In large systems it might be necessary to do this every week.)
- files from the Garbage folder.

5.3.2. Operator Console

If automatic deletion of TU and server events is turned off, delete old events manually. To delete events, select them in the relevant information view and press Delete on the keyboard.

It is however recommended to have automatic deletion of TU and server events turned on. See the Operator Console documentation regarding how to do this. Note that the stored procedures defined for this purpose in the database must first have been activated; see the Database Installation Guide for details.

6. References

- [1] TEMS Automatic Database Server Installation Guide
- [2] TEMS Automatic System Configuration and Dimensioning
- [3] SQL Server Books Online: Analysis Services\Administering Analysis Services\
Archiving, Restoring, and Copying Data
- [4] Microsoft Knowledge Base article 294287
- [5] SQL Server Books Online: Administering SQL Server\Backing Up and
Restoring Databases